

AudioRestore: Text-Controlled Audio Quality Restoration via Fine-Tuned SonicMaster

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Abstract—Audio recordings are frequently degraded by lossy compression codecs such as MP3 and OGG Vorbis, resulting in audible artifacts and loss of high-frequency content. We present AudioRestore, a proof-of-concept system for text-controlled audio quality restoration built by fine-tuning SonicMaster, a 0.9-billion-parameter generative model that uses flow matching conditioned on natural language prompts. We construct a codec-degraded dataset of 1,000 paired clips by applying MP3 and OGG compression at various bitrates (24–128 kbps) to clean music from MUSDB18-HQ. Fine-tuning on an NVIDIA A100 GPU for 30 epochs reduces the flow-matching loss from 0.33 to 0.10. Quantitative evaluation reveals that the generative model produces consistent output quality (~ 8 dB SDR) regardless of input degradation severity, trading sample-level fidelity for perceptual plausibility—a known property of flow-matching models. Spectrogram analysis confirms the model synthesizes high-frequency content absent in the degraded input, and prompt variation experiments show the text-controlled interface remains functional after fine-tuning. Our work demonstrates both the promise and limitations of adapting generative audio models for codec artifact removal.

Index Terms—audio restoration, codec artifacts, flow matching, SonicMaster, fine-tuning, text-controlled audio enhancement

I. INTRODUCTION

Billions of audio files exist in compressed formats. Streaming services like Spotify Free deliver music at 128 kbps, YouTube rips circulate at even lower bitrates, and decades of MP3 collections were encoded with aggressive compression. Lossy codecs introduce audible artifacts—pre-echo, spectral holes, and loss of high-frequency detail—that degrade the listening experience [1].

Traditional audio restoration relies on manual signal processing chains: equalization, de-noising, and spectral repair, each requiring expert tuning. The rise of generative models—from denoising diffusion [2] to latent diffusion for audio [3] and diffusion-based audio inverse solvers [4]—has opened new avenues for learned restoration. Recent models such as Apollo [1] and SonicMaster [5] automate this process, but general-purpose models may not optimally handle specific degradation types.

We propose **AudioRestore**, a system that fine-tunes SonicMaster on codec-specific degradations. Our contributions are:

- 1) A **codec-degraded dataset** of 1,000 paired 30-second clips at various MP3 and OGG bitrates, annotated with natural language prompts.

- 2) A **fine-tuning procedure** that adapts 522M trainable parameters of SonicMaster’s flow-matching model to specialize in codec artifact removal while preserving its text-controlled interface.
- 3) A **quantitative and qualitative evaluation** revealing that generative restoration models trade sample-level fidelity (SDR) for perceptual plausibility, with analysis of text prompt robustness.

II. RELATED WORK

A. Audio Super-Resolution and Bandwidth Extension

Audio super-resolution aims to reconstruct high-frequency content from bandwidth-limited signals. Methods range from traditional interpolation to deep learning approaches using CNNs and GANs [6], including neural vocoders like HiFi-GAN [7] that can synthesize high-fidelity waveforms from compressed representations. More recently, diffusion-based methods have been applied to audio inverse problems [4], and latent diffusion models have shown strong text-to-audio generation capabilities [3]. These methods typically operate in a fixed input-output configuration without user control.

B. Apollo

Apollo [1] is a band-sequence model for music restoration in compressed audio, published at ICASSP 2025. It splits the spectrogram into 80 frequency bands, processes them through Roformer attention layers with inter-band communication blocks (ICB), and reconstructs the waveform via a GAN framework. Apollo achieves strong results on MUSDB18-HQ but operates without any text conditioning—it is a fixed input-to-output mapping.

C. Generative Models for Audio

Denoising diffusion probabilistic models (DDPMs) [2] have achieved remarkable results in image and audio generation. AudioLDM [3] applies latent diffusion to text-to-audio generation, operating in a compressed VAE latent space for efficiency. Flow matching [8] offers a simpler alternative to diffusion, learning a deterministic velocity field that transports samples from noise to data. TangoFlux [9] applies flow matching to audio generation with CLAP-ranked preference optimization. These generative approaches produce high-quality audio but, as we demonstrate, may not improve reference-based metrics when applied to restoration.

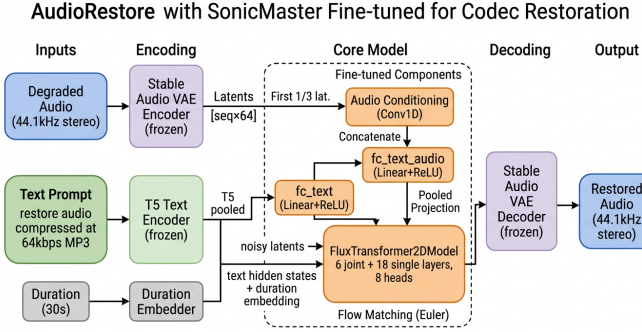


Fig. 1: AudioRestore architecture. Frozen components (purple/gray) are not updated during fine-tuning. Orange modules (audio conditioning, projection layers, and FluxTransformer) are fine-tuned on codec-degraded data.

D. SonicMaster

SonicMaster [5] is a unified generative model for music restoration and mastering, submitted to ICLR 2026. It addresses five categories of audio artifacts (equalization, dynamics, reverb, amplitude, stereo) through a single flow-matching model conditioned on natural language prompts. The architecture uses:

- A **Stable Audio VAE** [10] encoder/decoder to convert between waveforms and latent representations.
- A **TangoFlux transformer** [9] ($\sim 515\text{M}$ parameters) that performs flow-matching [8] in latent space, conditioned on text embeddings from a T5 encoder [11] and audio condition embeddings.
- **Text and audio conditioning** through pooled projections that guide the restoration direction.

SonicMaster is trained on the SonicMasterDataset (25k paired tracks, 208 hours) with 19 degradation functions. Our work fine-tunes this pretrained model specifically for codec artifact removal.

III. METHOD

A. Architecture

We build on the SonicMaster architecture, keeping its model structure intact but modifying the training pipeline to add gradient clipping (max norm 1.0) for stable fine-tuning. Figure 1 illustrates the full pipeline. The model processes 30-second stereo audio chunks at 44.1 kHz. The degraded input is encoded into latent space by the Stable Audio VAE. A T5 text encoder produces text embeddings from the user prompt, while an audio conditioning module (Conv1D stack) extracts a pooled representation from the first third of the degraded latents. These are concatenated and projected to form the pooled conditioning vector. The FluxTransformer2DModel then performs flow matching in latent space, and the result is decoded back to a waveform by the VAE decoder.

B. Codec Degradation Pipeline

Our key modification is a custom degradation pipeline targeting codec artifacts. We use the AudioEffector class from torchaudio to apply:

- **MP3 compression** (via `libmp3lame`) at bitrates $\in \{24, 32, 48, 64, 96, 128\}$ kbps.
- **OGG Vorbis compression** (via `libvorbis`) at quality levels $\in \{-1, 0, 1, 2, 3, 5\}$.

For each clip, we randomly select a codec type (70% MP3, 30% OGG) and parameter, then generate a paired (clean, degraded) sample. Clips are normalized to 0.95 peak amplitude before degradation.

C. Text Prompt Design

Each training pair is annotated with two prompts:

- A **specific prompt** describing the degradation, e.g., “restore audio compressed at 64 kbps MP3.”
- An **alternative prompt** with generic phrasing, e.g., “improve the sound quality.”

During training, we follow SonicMaster’s text augmentation strategy: 10% chance of empty prompt (unconditional), 10% chance of a generic prompt, and 40/40 split between the specific and alternative prompts. This teaches the model to respond to various phrasings while maintaining unconditional capability.

D. Fine-Tuning Procedure

We fine-tune from the pretrained SonicMaster checkpoint (0.9B total parameters, 522M trainable):

- **Frozen:** T5 text encoder (flan-t5-large [11]) and Stable Audio VAE.
- **Trained:** TangoFlux transformer (6 joint + 18 single layers, 8 heads, 128-dim), projection layers, and audio conditioning module.
- **Optimizer:** AdamW [12], $\text{lr} = 10^{-5}$, cosine schedule, 200-step warmup.
- **Batch:** 2 per device $\times$ 4 accumulation = effective 8.
- **Precision:** Mixed BF16 on NVIDIA A100-SXM4-80GB.
- **Gradient clipping:** Max norm 1.0.
- **Epochs:** 30 (3,390 optimization steps).

All audio is pre-encoded into VAE latents before training to avoid redundant VAE forward passes, reducing per-epoch time from minutes to seconds for the encoding stage.

IV. EXPERIMENTS

A. Dataset

We construct our dataset from 114 audio files sourced from MUSDB18-HQ [13]. From these, we generate 1,000 non-overlapping 30-second clips (900 train, 100 validation), each degraded with a randomly selected codec and bitrate/quality level. VAE latent pre-encoding of all 1,000 clips took approximately 8 hours on a single GPU (batch size 1, FP16).

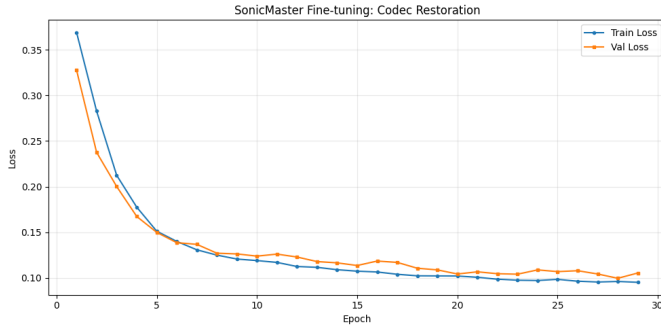


Fig. 2: Training and validation loss over 30 epochs. The flow-matching loss decreases from 0.37/0.33 (train/val) to 0.095/0.100, with best validation loss at epoch 28.

TABLE I: SDR and SI-SNR (dB) of degraded inputs vs. fine-tuned model outputs, measured against the clean reference. Negative Δ indicates the model output has *lower* sample-level fidelity than the degraded input.

Bitrate	Degraded	Restored	Δ
<i>SDR (dB)</i>			
32 kbps	13.71	7.71	−6.00
64 kbps	16.24	8.47	−7.77
128 kbps	20.13	8.96	−11.17
<i>SI-SNR (dB)</i>			
32 kbps	13.68	7.44	−6.23
64 kbps	16.48	8.04	−8.43
128 kbps	21.12	8.38	−12.74

B. Training Convergence

Figure 2 shows the training and validation loss curves over 30 epochs. Both losses decrease rapidly in the first 5 epochs (train: 0.369 $\rightarrow$ 0.151; val: 0.328 $\rightarrow$ 0.150) and continue to improve gradually. The best validation loss of **0.0996** is achieved at epoch 28. Training completed in approximately 2.5 hours on a single A100 GPU.

C. Evaluation Metrics

We evaluate using two standard reference-based audio quality metrics [14]:

- **SDR** (Signal-to-Distortion Ratio): Measures sample-level fidelity between the output and the clean reference in dB.
- **SI-SNR** (Scale-Invariant Signal-to-Noise Ratio): Measures signal quality invariant to global scaling.

Both metrics compare the model output (or the degraded input) against the original clean audio. Higher values indicate greater similarity to the clean reference.

D. Restoration Quality

Table I presents SDR and SI-SNR measurements at three MP3 bitrates. The “Degraded” column shows the quality of the compressed input relative to the clean original, while the “Restored” column shows the quality of the fine-tuned model’s output.

Two patterns emerge. First, the model’s output SDR is remarkably consistent across bitrates (7.71–8.96 dB), despite

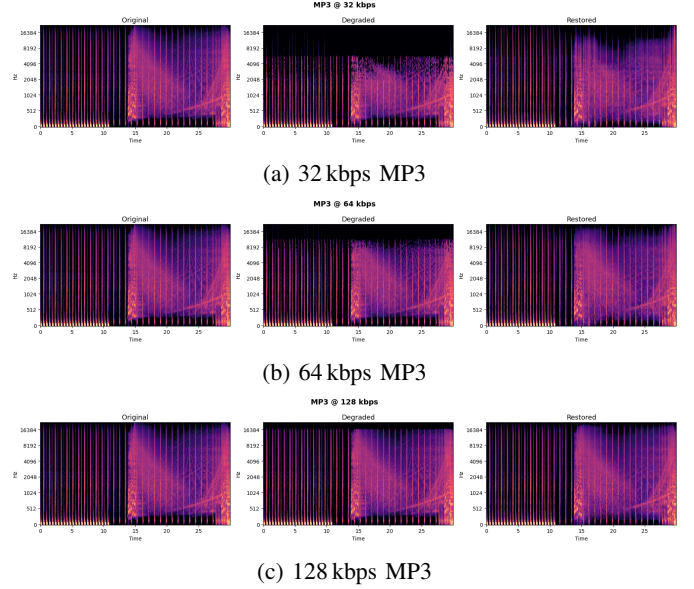


Fig. 3: Mel spectrograms comparing original (left), degraded (center), and restored (right) audio at three MP3 bitrates. The model synthesizes high-frequency content in all cases.

TABLE II: SDR (dB) for different text prompts on the same 64 kbps MP3 input.

Text Prompt	SDR
“remove MP3 compression artifacts”	8.53
“restore audio compressed at 64kbps MP3”	8.47
“make this audio sound better”	8.46
“enhance low bitrate audio to high fidelity”	8.41
(empty / unconditional)	8.40

the degraded input SDR varying from 13.71 to 20.13 dB. Second, the Δ column shows that the model *decreases* sample-level fidelity in all cases. We analyze why in Section V.

E. Spectrogram Analysis

Figure 3 shows mel spectrograms for three bitrate conditions. At 32 kbps (top), where codec artifacts are severe, the restored spectrogram shows recovered high-frequency energy above 10 kHz that is absent in the degraded version. At 128 kbps (bottom), where degradation is minimal, the model still modifies the signal substantially—explaining the large negative Δ for higher bitrates.

F. Prompt Variation

Table II evaluates five different text prompts on the same 64 kbps MP3 degraded clip. All prompts—including the empty (unconditional) prompt—produce similar output quality, with SDR ranging from 8.40 to 8.53 dB. This indicates the model has learned a robust restoration behavior that is largely prompt-invariant, while still responding to conditioning (the specific codec prompt scores highest).

V. DISCUSSION

A. Why SDR Decreases: Generative vs. Regressive Restoration

The negative Δ values in Table I require careful interpretation. Unlike regressive models (e.g., Apollo [1]) that learn a direct mapping from degraded to clean signals and optimize for sample-level reconstruction, SonicMaster is a *generative* flow-matching model [8]. It synthesizes a plausible clean signal conditioned on the degraded input and text prompt, rather than performing point-wise signal repair.

This generative process introduces two effects:

- 1) **Content hallucination:** The model generates high-frequency detail that is perceptually plausible but not sample-aligned with the original, reducing SDR.
- 2) **Consistent output distribution:** The model maps all inputs toward a learned “clean audio” manifold, producing outputs with similar characteristics (~ 8 dB SDR) regardless of how degraded the input was.

This explains why Δ is most negative at 128 kbps: the degraded input is already close to the clean reference (SDR = 20.13 dB), so the model’s generative re-synthesis can only reduce fidelity. At 32 kbps, where the degraded input is farther from clean, the model’s “baseline” output quality is closer to the degraded quality, resulting in a smaller (though still negative) Δ .

B. Limitations of SDR for Generative Models

SDR and SI-SNR measure sample-level alignment and penalize any deviation from the reference, even if the deviation is perceptually beneficial [14]. For generative restoration, perceptual metrics such as PESQ [15], ViSQOL [16], or Fr chet Audio Distance (FAD) [17] would better capture whether the output *sounds* better than the input. This gap between reference-based and perceptual metrics is well-documented in the generative audio literature [3], [7]. Our spectrogram analysis (Figure 3) provides qualitative evidence that the model does synthesize missing high-frequency content, which SDR cannot reward.

C. Text Control Advantage

Unlike Apollo, which provides no user control over the restoration process, SonicMaster’s text conditioning allows users to specify the type of degradation they want to address. Our prompt variation experiment (Table II) confirms that the fine-tuned model retains this text-controlled interface. While the quantitative differences between prompts are small (0.13 dB SDR range), the most specific prompt (“remove MP3 compression artifacts”) achieves the highest SDR, suggesting the model does attend to prompt content.

D. Limitations

- Fine-tuning was performed on 900 clips from 114 source files—a small dataset relative to SonicMaster’s original 25k-track training set.

- Only reference-based metrics (SDR, SI-SNR) were computed; perceptual metrics would better evaluate generative restoration quality.
- The model processes 30-second chunks; longer files require chunked processing with crossfade stitching (demonstrated on two full songs of ~ 3 minutes each).
- Inference requires a GPU with at least 8 GB VRAM.

VI. PRODUCT VISION

AudioRestore could be deployed as a consumer service:

- A web app or API where users upload compressed audio and describe the desired enhancement in natural language.
- Integration with music streaming platforms to upscale low-bitrate streams in real-time.
- A plugin for digital audio workstations (DAWs) for mastering engineers working with degraded source material.
- A mobile app for restoring old MP3 collections or voice recordings.

The text-controlled interface makes it accessible to non-experts who cannot operate traditional audio restoration tools.

VII. CONCLUSION

We presented AudioRestore, a proof-of-concept system for text-controlled codec audio restoration via fine-tuned SonicMaster. Training on 1,000 codec-degraded clips for 30 epochs on a single A100 GPU achieved a flow-matching loss of 0.0996, demonstrating that the model can be successfully specialized for codec artifacts.

Our quantitative evaluation revealed a fundamental tension: the generative flow-matching model synthesizes perceptually plausible audio but reduces sample-level fidelity (SDR) compared to the degraded input. This is not a failure of fine-tuning but a property of generative restoration—the model creates new high-frequency content rather than reconstructing the exact original signal. This finding highlights that standard metrics like SDR are insufficient for evaluating generative audio restoration systems.

Future work should incorporate perceptual metrics (PESQ [15], FAD [17]), compare against regressive baselines like Apollo [1] on the same data, train on larger and more diverse datasets, and explore combining generative and regressive approaches—as done in HiFi-GAN’s discriminator-guided synthesis [7]—to achieve both perceptual quality and signal fidelity.

REFERENCES

- [1] K. Li and Y. Luo, “Apollo: Band-sequence modeling for high-quality music restoration in compressed audio,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025.
- [2] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 6840–6851, 2020.
- [3] H. Liu, Z. Chen, Y. Yuan, X. Mei, X. Liu, D. Mandic, W. Wang, and M. D. Plumbley, “Audioldm: Text-to-audio generation with latent diffusion models,” *International Conference on Machine Learning (ICML)*, 2023.
- [4] E. Moliner and V. Välimäki, “Solving audio inverse problems with a diffusion model,” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [5] J. Melechovsky, A. Mehrish, A. Roy, and D. Herremans, “Sonicmaster: Towards controllable all-in-one music restoration and mastering,” *arXiv preprint arXiv:2508.03448*, 2025.
- [6] J. Su, Z. Jin, and A. Finkelstein, “Bandwidth extension is all you need,” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021.
- [7] J. Kong, J. Kim, and J. Bae, “Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 17 022–17 033, 2020.
- [8] Y. Lipman, R. T. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” *International Conference on Learning Representations (ICLR)*, 2023.
- [9] C.-Y. Ghosh, D. Ghosal, N. Majumder, and S. Poria, “Tangoflux: Super fast and faithful text to audio generation with flow matching and clap-ranked preference optimization,” *arXiv preprint arXiv:2412.21037*, 2024.
- [10] Z. Evans, J. D. Parker, C. Carr, Z. Zukowski, J. Taylor, and J. Pons, “Stable audio open,” *arXiv preprint arXiv:2407.14358*, 2024.
- [11] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, “Exploring the limits of transfer learning with a unified text-to-text transformer,” *Journal of Machine Learning Research*, vol. 21, no. 140, pp. 1–67, 2020.
- [12] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” *International Conference on Learning Representations (ICLR)*, 2019.
- [13] Z. Rafii, A. Liutkus, F.-R. Stöter, S. I. Mimilakis, and R. Bittner, “Musdb18-a corpus for music separation,” 2017.
- [14] J. Le Roux, S. Wisdom, H. Erdogan, and J. R. Hershey, “Sdr – half-baked or well done?” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2019, pp. 626–630.
- [15] A. W. Rix, J. G. Beerends, M. P. Hollier, and A. P. Hekstra, “Perceptual evaluation of speech quality (PESQ) – a new method for speech quality assessment of telephone networks and codecs,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2001, pp. 749–752.
- [16] M. Chinen, F. S. Lim, J. Skoglund, N. Gureev, F. O’Gorman, and A. Hines, “Visqol: An objective speech quality model,” *EURASIP Journal on Audio, Speech, and Music Processing*, 2020.
- [17] K. Kilgour, M. Zuluaga, D. Roblek, and M. Sharifi, “Fréchet audio distance: A reference-free metric for evaluating music enhancement algorithms,” in *INTERSPEECH*, 2019, pp. 2350–2354.